Let's make protein

Every cell in your body has DNA. The DNA is tightly packed and coiled in the nucleus of each cell. A section of DNA is called a gene. This gene is made up of a sequence of bases that code for a protein. Your body continually makes protein for growth and repair. During **protein synthesis**, only part of the DNA is read – the rest is 'switched off'.

The following activity will help you understand the complexity of protein synthesis. The process of protein synthesis has been written in a procedure format. Carefully read the procedure and complete the activity.

The process of protein synthesis

Aim: To simulate how protein is made.

Materials needed:

- strand of DNA, about one gene
- one messenger RNA molecule
- one ribosome
- many transfer RNA molecules

Method:

- Study the information below and the diagrams on the next page, which show the steps involved in protein synthesis in the human body.
- Cut out the steps and the diagrams.
- Arrange the diagrams in order from 1 to 8. Then match each information box to the correct diagram.
- Check your work with a classmate. When you agree on the order, paste the steps and diagrams in your workbook under the heading 'Protein synthesis'.

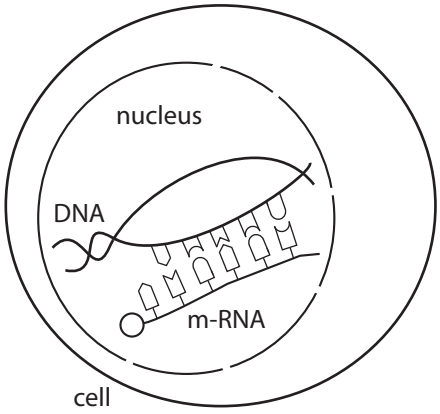
Extension

In your workbook, rewrite this procedure as a scientific explanation for how protein is synthesised in human cells. Use the procedure you have just ordered as a guide.

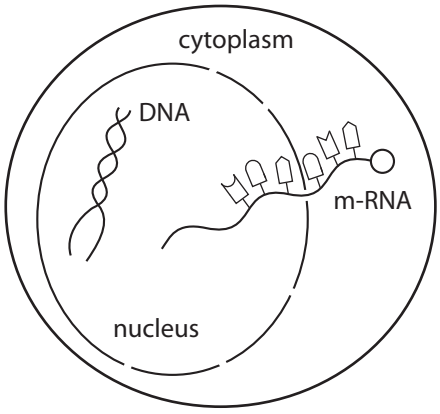
a Collect the transfer RNA molecules needed.	**b** Move the messenger RNA out of the nucleus into the cytoplasm.
c Add the second t-RNA onto the m-RNA and join each new amino acid to the previous one. Remove the first t-RNA as soon as the amino acid is connected to the second amino acid.	**d** Unzip the piece of DNA. Copy the code from one strand onto the messenger RNA by matching the bases adenine with thymine, and guanine with cytosine. (*Hint:* RNA uses uracil instead of thymine.)
e Connect the m-RNA onto a ribosome where the proteins will be assembled.	**f** Connect the t-RNA one at a time onto the m-RNA, matching to the triplet codes – three bases m-RNA.
g The completed protein should fall off.	**h** Repeat steps 5 and 6 for 200 amino acids until the stop code is reached on the m-RNA.

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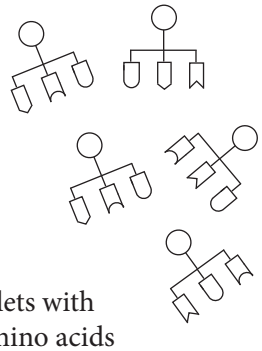
Let's make protein



1

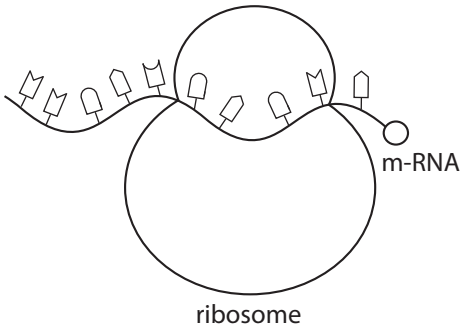


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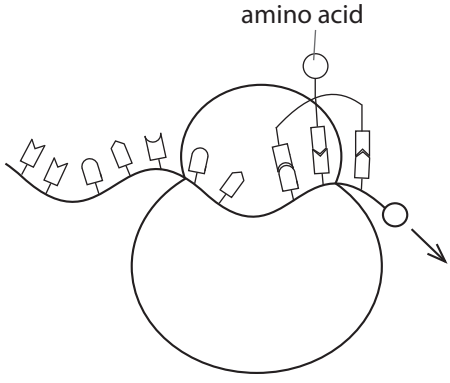


t-RNA triplets with attached amino acids

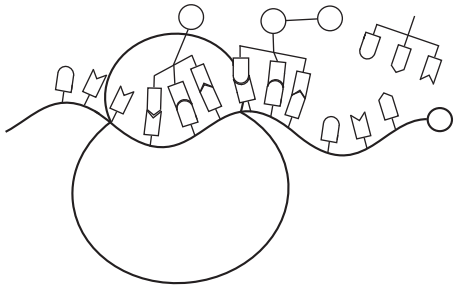
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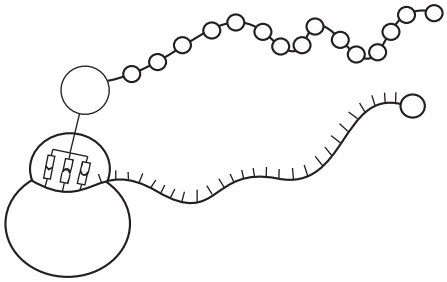
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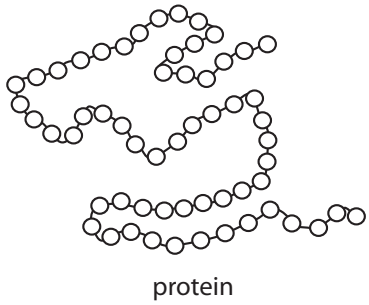
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6



7



8

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